



SECP2523: DATABASE

SEM I 2024/2025

DATABASE DESIGN DESCRIPTIONS: CONCEPTUAL DATABASE DESIGN FOR KADA ESERVE

Group Name: TECH AWAY

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Stakeholder: Lembaga Kemajuan Pertanian Kemubu (KADA)

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TABLE OF CONTENTS

CHAPTER	TOPIC	PAGE
1	Introduction	
1.1	Overview about the company	Page 3
1.2	Problems	Page 3
1.2	Proposal	Page 4
1.3	Definitions, Acronyms and Abbreviations	Page 5
2	Database System Requirement	
2.1	System Overview	Page 6
2.2	Module 1 <User Module>	Page 7-8
2.2.1	Module Description	
2.2.2	Transaction Requirement: Data Entry	
2.2.3	Transaction Requirement: Data Update/Deletion	
2.2.4	Transaction Requirement: Data Queries	
2.2.5	Local Conceptual Data Model (ERD) for Module 1	
2.3	Module 2 <Member Module>	Page 9-11
2.3.1	Module Description	
2.3.2	Transaction Requirement: Data Entry	
2.3.3	Transaction Requirement: Data Update/Deletion	

2.3.4	Transaction Requirement: Data Queries	
2.3.5	Local Conceptual Data Model (ERD) for Module 2	
2.4	Module 3<Clerk Module>	Page 11-15
2.4.1	Module Description	
2.4.2	Transaction Requirement: Data Entry	
2.4.3	Transaction Requirement: Data Update/Deletion	
2.4.4	Transaction Requirement: Data Queries	
2.4.5	Local Conceptual Data Model (ERD) for Module 3	
3	Global Conceptual Data Model (Global ERD)	Page 16-27
3.1	Global Conceptual Design (ERD)	
3.2	Verified Global Conceptual Design (Verified ERD)	
3.2	Data Dictionary for the global conceptual ERD.	
4	Appendices	Page 28-33
4.1	Use Case Diagram for the system.	
4.2	Activity Diagrams for every use case identified.	
4.3	Meeting Log	

NOTE: A module can consist of one or multiple (related) use case. A team member in a group must be responsible to at least one module.

1. Introduction

1.1 Overview about the company

The Lembaga Kemajuan Pertanian Kemubu(KADA) is a Malaysian government agency under the Ministry of Agriculture and Food Security. KADA's primary objective is to spearhead the development of the agriculture sector in the Kemubu area, a key agricultural region in Kelantan. KADA is dedicated to improving the socioeconomic status of farmers through modern farming techniques, infrastructure development, and effective management of resources.

1.2 Problems

KADA currently relies heavily on manual, paper-based processes for member registration, loan applications, and data management. This traditional approach has led to several challenges that hinder operational efficiency and the organization's ability to adapt to modern demands.

Inefficiency in Member Registration and Data Entry

The manual registration process requires users to fill out paper forms, which requiring staff must then process and input into the system manually. This process is time-consuming and prone to errors, such as incorrect data input or duplication of records. These inefficiencies result in inconsistent data integrity and delays in updating the database, which affects overall system responsiveness and member onboarding timelines.

Limited Data Storage Capacity

KADA's current digital storage capacity for member registration is limited to on a few hundred members, posing a significant obstacle to the organization's expansion. The inability to handle a larger volume of data restricts KADA's ability to recruit and manage more members. This limitation also affects the organization's capacity to maintain comprehensive records, generate detailed reports, and conduct effective data analysis, all of which are vital for strategic planning and operational efficiency.

Security

The reliance on manual data entry and paper forms in KADA's current system presents significant security risks. Without proper encryption and access controls, the database is exposed to vulnerabilities, and manual entry increases the errors such as incorrect or incomplete records. These gaps compromise data confidentiality and operational efficiency.

Lack of Resources to Support Decision Making

The current database system lacks the features necessary to support advanced data analytics and reporting. Data is fragmented and stored in an unstructured format, hindering the ability to generate insightful reports and monitor operational trends. This limitation affects KADA's capacity to leverage historical data for strategic decision-making and hampers its ability to perform predictive analysis for future planning.

1.3 Proposal

The system will be developed and named KADA eServe, a digital solution designed to streamline the operational process of KADA while enhancing user experience. This system has to meet the organization's needs and provides efficient management of user registration, loan applications and administrative.

To address the challenges posed by KADA's reliance on manual processes and limited digital systems, we propose the implementation of a robust, scalable and secure database-driven solution. This proposal outlines how the adoption of a database system can resolve the identified inefficiencies, improve operational efficiency and support strategic growth.

For the first problem that the company faced is addressing inefficiency in member registration and data entry. The inefficiency in member registration can be resolved by implementing an online registration system integrated with a centralized database. Members can complete digital registration forms which are validated and stored directly in the database without the need for manual data entry. Validation rules will also ensure that all inputs are consistent and error-free, reducing the chances of incorrect or duplicate records. By automating the data entry process, this solution streamlines member onboarding, making member data immediately available for other system functionalities.

Second, to overcome the limitations of KADA's current storage capacity, we have implemented a scalable relational database management system (RDBMS) such as MySQL is proposed. This will enable the system to effectively handle thousands of member records and support future expansion. Storage optimisation can be done by normalising data structures which ensuring consistency and cuts down on redundancy. Additionally, a data archiving strategy will ensure that inactive or historical records are stored securely without affecting system performance.

Next, the security risks associated with manual processes can be mitigated by integrating robust security mechanisms into the database system. AES and SSL/TLS encryption protocols will be used to protect sensitive data while it is being transmitted and while it is at rest. To improve confidentiality, role-based access restrictions will make sure that only individuals with permission can view or alter particular data. By taking these steps, the system's security will be greatly improved and member data will be shielded from breaches.

Lastly, a structured relational database will enable efficient querying and analysis of data. Real-time reporting will provide insights into member activities, loan statuses, and organizational performance, empowering administrators to monitor trends effectively.

In conclusion, by implementing these database solutions, KADA can transition to a modern and efficient system that overcomes existing challenges, enhances security, supports scalability and provides actionable insights for informed decision-making.

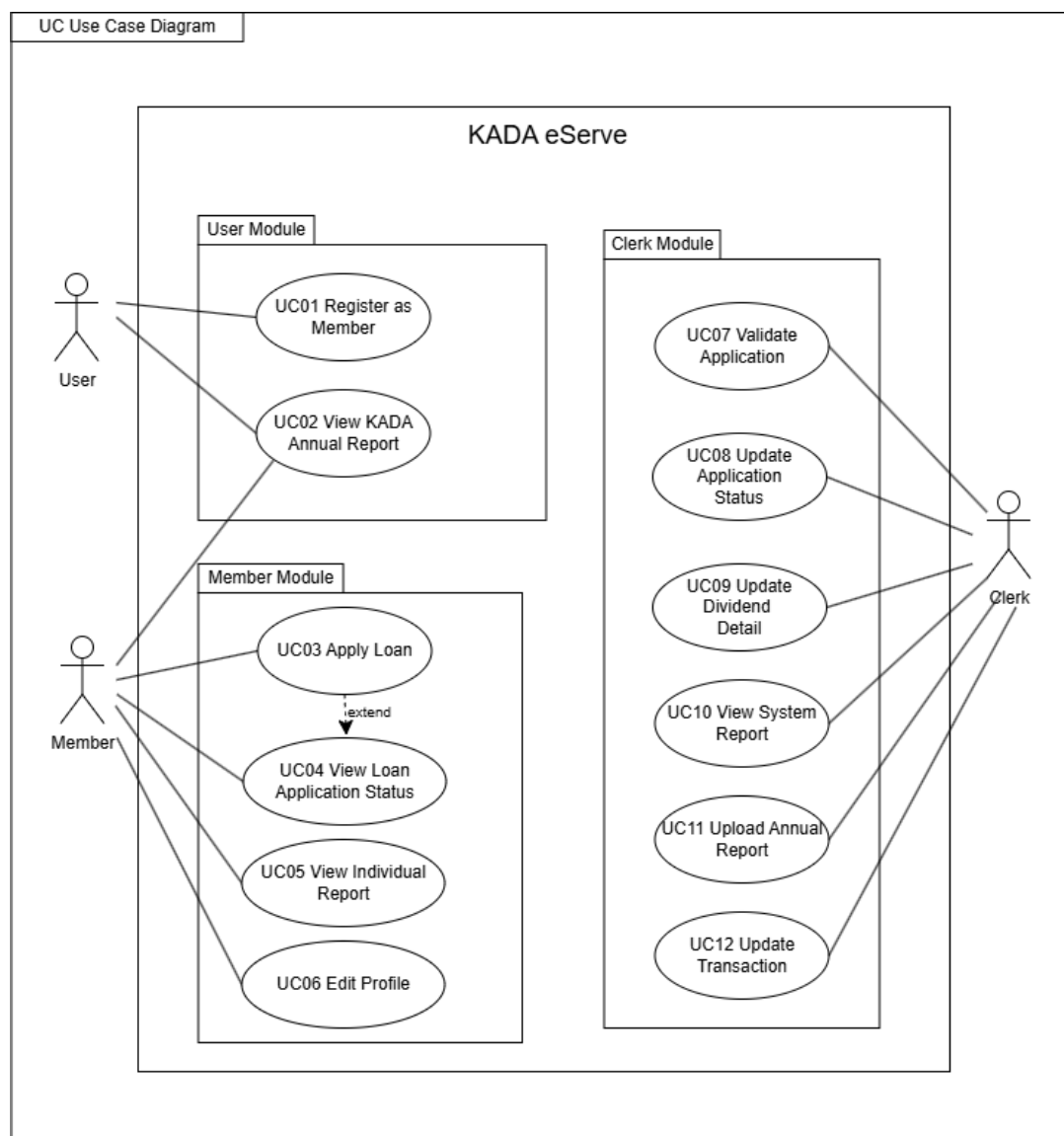
1.4 Definitions, Acronyms and Abbreviations

No.	Acronyms/Abbreviations	Definition
1.	User	An individual that not registered as KADA system's member, able to register as member and view annual report of the company
2.	Member	An individual registered with the KADA system, eligible to apply for loans and view individual reports.
3.	Clerk	A system administrator that responsible for validating applications, updating application status and managing dividend details.
4.	KADA	Lembaga Kemajuan Pertanian Kemubu ((Kemubu Agricultural Development Authority)
5.	UML	Unified Modeling Language: A standardized modeling language used to visualize the design of a system.
6.	UC	Use Case, representing specific user interactions with the system
7.	ERD	Entity-Relationship Diagram, a visual tool used to model the structure of a database.

2. Database System Requirement

2.1 System Overview

The KADA System is a document management system (DMS) which able to solve problem faced by KADA currently. The main purpose of this system is to digitize and automate the flow of member registration, loan application and reporting for the KADA member as the current management system, paper-based document management system is less efficiency. Otherwise, transition to digitize system not only helps to improve the efficiency of workflows but also ensuring the data security for protecting user's information and improve user convenience. In another word, this system is a critical step toward transforming KADA into a modern, efficient, and technology-driven organization.



2.2 Module 1 <User Module>

Tan Zhi Ming is responsible for documenting this module.

2.2.1 Module Description

This module will involve only one actor: Users. It includes two use cases which is:

1. UC01: Register as a Member
2. UC02: View the KADA Annual Report

Register as a Member (UC01)

- The system shall provide a dedicated registration page accessible to visitors.
- The system shall display registration form
- The system shall validate all form inputs to ensure completeness and correctness.
- The system shall store the registration details in the local database upon success validation.
- The system shall generate a unique Member ID for each registered user.
- The system shall notify user of successful submission with a confirmation message.
- The system shall display error messages for invalid input or duplicate records.

View the KADA Annual Report (UC02)

- The system shall provide a dedicated section on the website for users to access the annual report.
- The system shall display a list of available annual reports, including metadata such as year and title.
- The system shall display a user-friendly error message, If the requested report is not available.
- The system shall allow the administrator to upload the new annual report through the security management panel.

2.2.2 Transaction Requirement: Data Entry

1. **Register as a Member (UC01):** KADA staff enter their information, set a password, and submit the application. They then wait for approval to become a member.

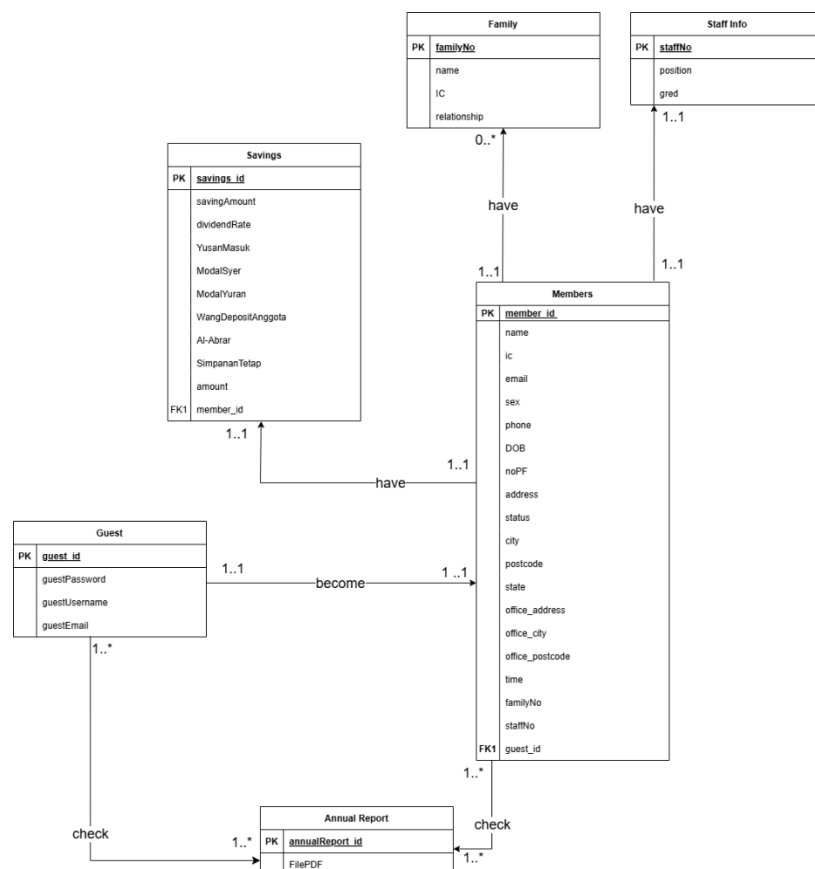
2.2.3 Transaction Requirement: Data Update/Deletion

No data update or deletion are included in this module

2.2.4 Transaction Requirement: Data Queries

1. **Register as a Member (UC01):** List of user's email to ensure no duplicate email address can be used in register.
2. **View the KADA Annual Report (UC02):** Retrieve the PDFs of KADA annual reports.

2.2.5 Local Conceptual Data Model (ERD) for Module 1



2.3 Module 2 <Member Module>

Brendan Chia Yan Fei is responsible for documenting this module

2.3.1 Module Description

The actor involved in this module is KADA member while the use cases included in this module are:

1. UC03 Apply Loan
2. UC04 View Loan Application Status
3. UC05 View Individual Report
4. UC06 Edit Profile

UC03 Apply Loan

- The system shall provide a “Loan Application” button in the dropdown menu
- The system shall allow members to enter the desired loan amount, preferred interest rate and loan repayment period
- The system shall display the monthly payment schedule upon approval.
- The system shall display the calculated details to the member.
- The system shall store the loan repayment details in database for tracking
- The system should notify the clerk of pending loan applications for verification and allow the clerk to approve or reject loan applications.

UC04 View Loan Application Status

- The system should provide an interface for members to view loan application status after applying for a loan.
- The system shall display one of the three statuses to the loan applicant, whether approved, pending or rejected.

UC05 View Individual Report

- The system shall provide a “Individual Report” button in the dropdown menu.
- The system shall display the details of the individual report page (e.g. Member details, Savings, Total Loan, Dividends, Loan Repayment Dates, Loan Application Success Dates, Transaction).
- The system shall display a user-friendly error message if the requested visitor is a non-KADA member.

UC06 Edit Profile

- The system shall provide a “Edit Profile” button in the profile page.
- The system shall allow the user to edit his/her personal information which they submitted through the form to register as a KADA member
- The system shall provide a “Save” button at the bottom right to pass the updated data to the database

2.3.2 Transaction Requirement: Data Entry

1. **Apply Loan (UC03):** KADA member enter their preferred loan type, loan term, loan amount, monthly installment and their personal details

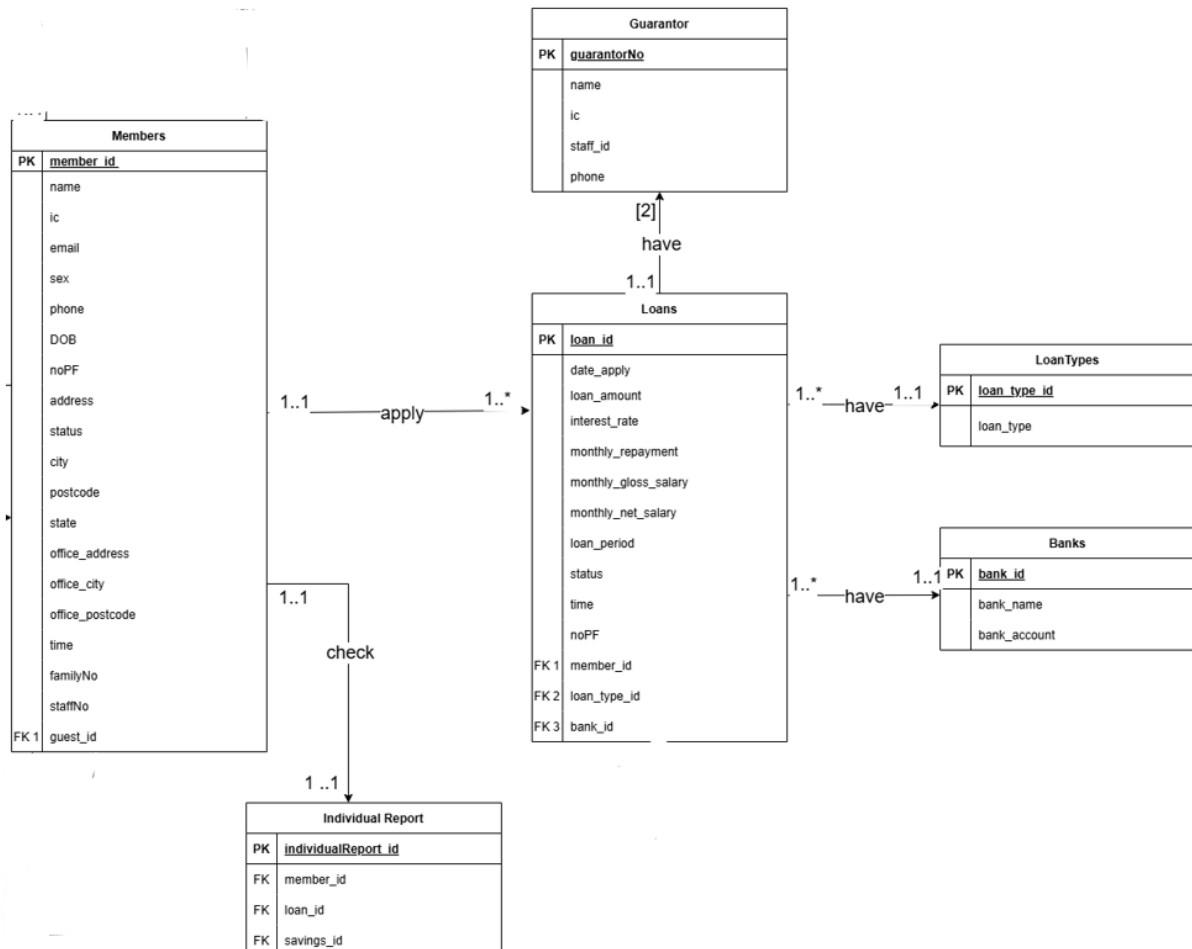
2.3.3 Transaction Requirement: Data Update/Deletion

1. **Edit Profile (UC06):** If validation succeeds, the system will update the user’s personal information in the database with the new data

2.3.4 Transaction Requirement: Data Queries

1. **View Individual Report (UC05):** Retrieve the individual report of KADA members which displays Member details, Savings, Total Loan, Dividends, Loan Repayment Dates, Loan Application Success Dates, Transaction
2. **View Loan Application Status (UC04):** Retrieve the status of loan application (Pending/Approved/Rejected)

2.3.5 Local Conceptual Data Model (ERD) for Module 2



2.4 Module 3 <Clerk Module>

This module will only be handled by Gui Kah Sin.

2.4.1 Module Description

The actor involved in this module is KADA clerk, while the use cases will be:

1. UC07 Validate Application
2. UC08 Update Application Status
3. UC09 Update Dividend Details
4. UC10 View System Report
5. UC11 Update Annual Report
6. UC12 Update Transaction

UC07 Validate Application

- The system shall provide a "Validate Applications" button in the navigation bar.
- The system shall display a list of pending membership and loan applications when the clerk clicks the "Validate Applications" button.
- The system shall allow the clerk to review the application details to ensure the accuracy and completeness of information.
- The system shall allow the clerk to mark the application as:
 - **Validated** if all criteria are met.
 - **Rejected** if missing information or documents are identified, with a comment field for specifying the reason for rejection.
- The system shall allow the clerk to resume the validation process once the applicant resubmits their complete application.
- The system shall notify the clerk if a technical error occurs during the validation process.
- The system shall store a record of the validation activity for auditing and tracking purposes.

UC08 Update Application Status

- The system shall provide an "Update Application Status" button in the navigation bar.
- The system shall display a list of applications with their current statuses (e.g., Pending, Validated, Rejected).
- The system shall require the clerk to provide comments when updating the application status to "Rejected."
- The system shall prompt the clerk to enter comments if they attempt to select "Rejected" without providing comments.
- The system shall update the application status in the database upon submission.
- The system shall notify the applicant about the status update and the next steps if the status is "Rejected" via an e-mail.
- The system shall store a record of the status update in the system for auditing and tracking purposes.

UC09 Update Dividend Details

- The system shall provide an "Update dividend details" button in the navigation bar.
- The system shall provide a column for updating the dividend rate.
- The system shall verify the entered dividend rate to avoid an illogical value being updated, such as a negative value or an extremely large value.

- The system shall prompt the clerk to decide whether the entered rate is correct before it is successfully updated to the whole system.
- The system shall notify the clerk if the update is unsuccessful or the update rate is illogical.
- The system shall store a record of the latest dividend details in the system for calculating savings for members.
- The system shall update the dividend details and savings of members in their own individual report.

UC10 View System Report

- The system shall provide a summary of member registrations, including the success and failure rates, along with demographic details.
- The system shall offer insights into the number and status of loan applications, approval or rejection rates, and related financial summaries.
- The system shall display data on member savings, including the total savings, dividend details, and *growth trends*.
- The system shall analyze and present savings trends, showing year-over-year in member savings behavior.
- The system shall allow clerk to upload and delete annual report PDFs.
- The system shall be able to view data export in formats like Excel or PDF for external analysis or reporting purposes.

UC11 Update Annual Report

- The system shall clerk to allow upload annual report PDF file.
- The system shall validate the file format and size before accepting the upload.
- The system shall notify the clerk if the upload is successful or if an error occurs.
- The system shall allow clerk to delete outdated or incorrect annual report PDF file.
- The system shall prompt the clerk for confirmation before deletion.

UC12 Update Transaction

- The system shall query the database for members with approved applications.
- The system shall display a list of members in ascending order, including staff name and staff ID.
- The system shall allow the “Add” function for the clerk to update either the member loan repayment or savings record.
- The system shall notify the clerk whether the transaction is a success or an error.

2.4.2 Transaction Requirement: Data Entry

1. **Validate Application (UC07):** KADA clerk will enter comments when rejecting applications to notify the error of members and record validation activities for any future purposes.
2. **Update Dividend Details (UC09):** KADA clerk will update dividend rates.
3. **View System Report (UC10):** Any new data processed or entered reflects in the system report.
4. **Update Annual Report (UC11):** KADA clerk upload the annual report PDF, with validation of file format and size during the entry process.
5. **Update Transaction (UC12):** KADA clerk will add loan repayment or savings records for approved members.

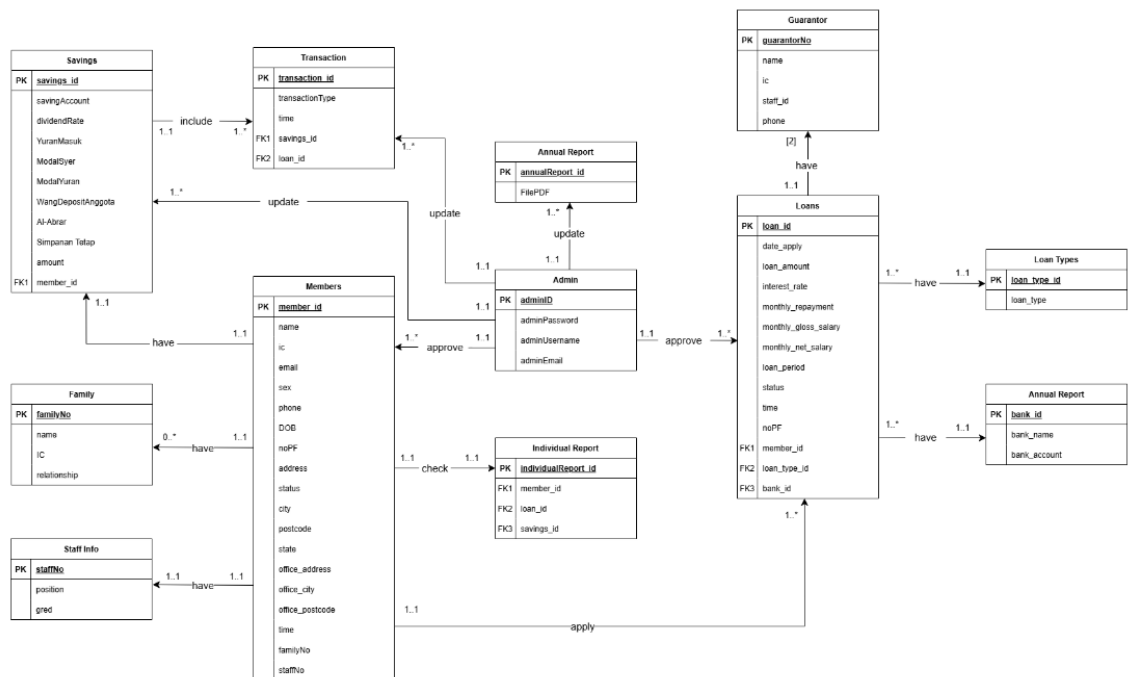
2.4.3 Transaction Requirement: Data Update/Deletion

1. **Validate Application (UC07):** KADA clerk will update the status of applications to “Validated” or “Rejected”. They will mark resubmitted applications for future review.
2. **Update Application Status (UC08):** KADA clerk will update the application status in the database and attaching comments for those who being rejected.
3. **Update Dividend Details (UC09):** KADA clerk will modify dividend rates and ensuring they propagate across the system to member individual reports.
4. **Update Annual Report (UC11):** KADA clerk can delete outdated or incorrect annual reports, with a confirmation prompt before deletion.
5. **Update Transaction (UC12):** KADA clerk updates members’ loan repayment or savings records, ensuring transaction data accuracy.

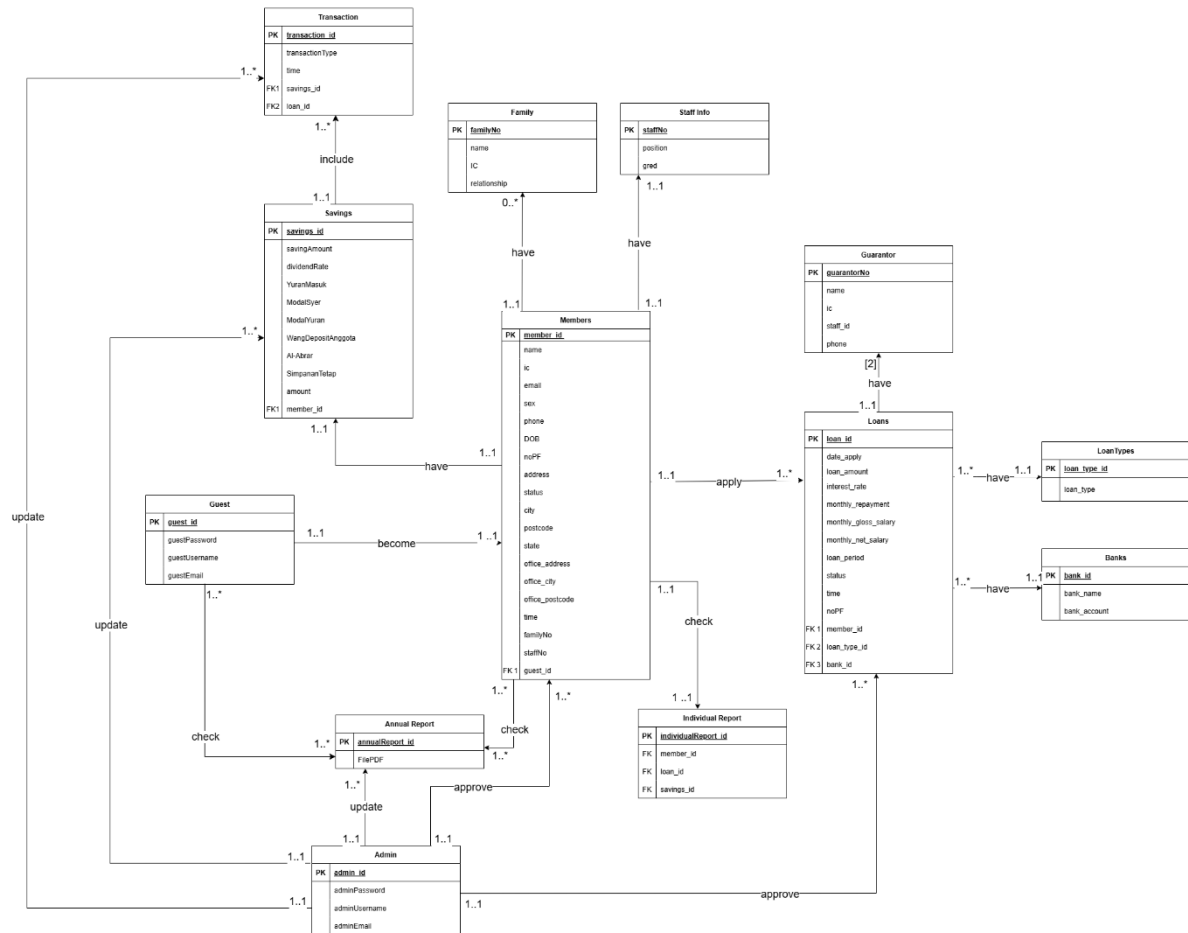
2.4.4 Transaction Requirement: Data Queries

1. **Validate Application (UC07):** KADA clerk will query a list of pending applications for validation.
2. **Update Application Status (UC08):** KADA clerk will retrieve applications with their current statuses.
3. **View System Report (UC10):** System will generate summaries and statistics for member registrations, savings, and loan applications.
4. **Update Annual Report (UC11):** KADA clerk will query and retrieve existing annual reports stored in the system.
5. **Update Transaction (UC12):** KADA clerk will query a list of members with approved applications, displayed in ascending order by name, along with their staff IDs.

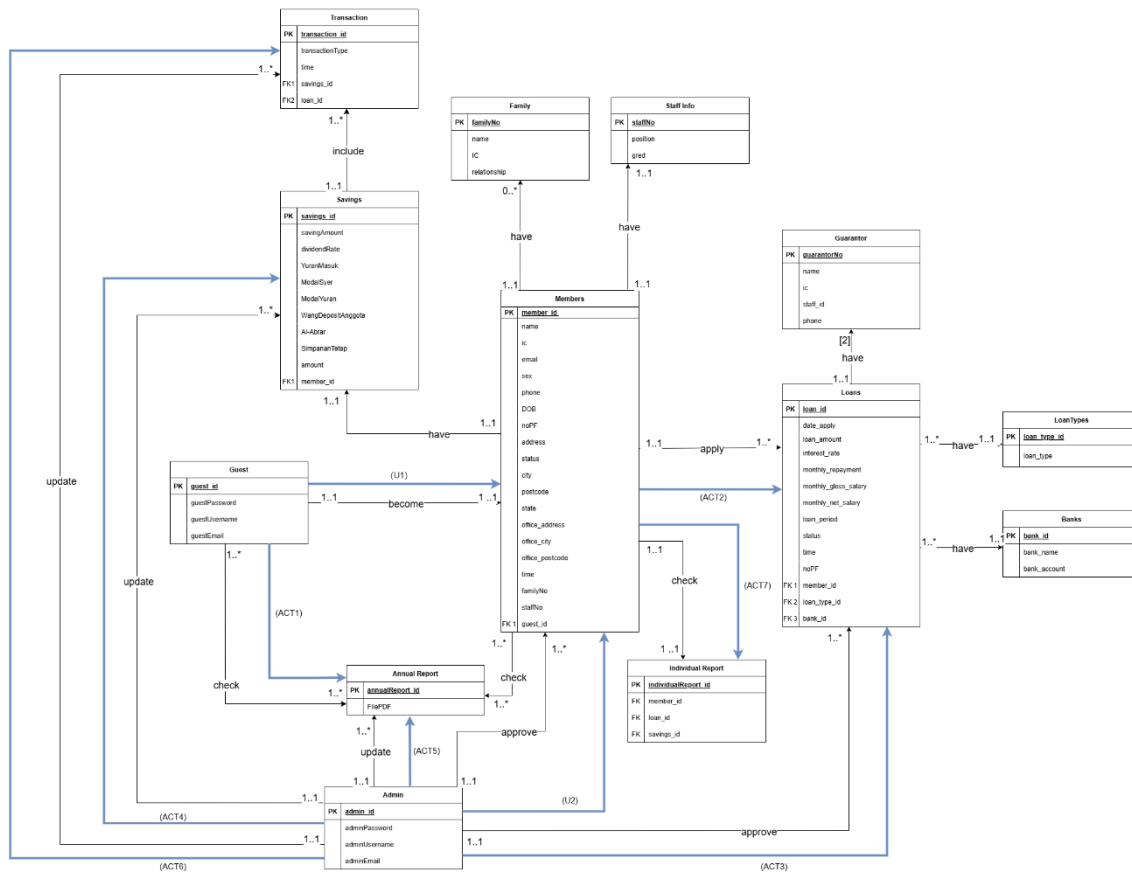
2.4.5 Local Conceptual Data Model (ERD) for Module 3



3.1 Global Conceptual Design (ERD)



3.2 Verified Global Conceptual Design (Verified ERD)



Symbol	Transaction
U1	2.2.2.1, 2.2.4.1, 2.3.3.1
U2	2.4.2.3, 2.4.4.3
ACT1	2.2.4.2
ACT2	2.3.2.1
ACT3	2.4.2.1, 2.4.2.3, 2.4.3.1, 2.4.3.2, 2.4.3.5, 2.4.4.1, 2.4.4.2, 2.4.4.3
ACT4	2.4.2.2, 2.4.2.3, 2.4.3.3, 2.4.3.5, 2.4.4.3
ACT5	2.4.2.4, 2.4.3.4, 2.4.4.4
ACT6	2.4.2.5, 2.4.3.5, 2.4.4.5
ACT7	2.3.4.1

3.3 Data Dictionary for the global conceptual ERD

3.3.1 Identifying Entity

Entity Name	Description	Aliases	Occurrence
Guest	General term describing all users KADA-eServe	Visitor	Occurs when a user register in the system
Member	General term describing all user logged in as member using KADA-eServe	Registered staff	Occurs when a guest registration application approved by admin
Admin	General term describing admin managing KADA-eServe	Clerk, Administrator, manager	Occurs when the system built
Transaction	General term describing all transaction record key-in by admin	Deposit amount, loan repayment	Occurs when admin key-in the amount of deposit or loan repayment for each member
Savings	General term describing the savings info of each member	Yuran, Modal, Deposit, Simpanan	Occurs when a guest registration application approved by admin
Family	General term describing the family info of each member	Pewaris, Keluarga	Occurs when a guest registration application approved by admin
Staff Info	General term describing the staff position, grade info	KADA staff Info	Occurs when a guest registration application approved by admin
Guarantor	General term describing the guarantor of the member	Warrantor	Occurs when a guest registration application approved by admin
Loan	General term describing the loan info of member	Pembiayaan	Occurs when a member submits the loan application form to system

LoanTypes	General term describing the loan info of member	Loan type	Occurs when a member submits the loan application form to system
Banks	General term describing the bank account info of member	Bank account	Occurs when a member submits the loan application form to system
Individual Report	General term describing the individual report of each member saving and loan info.	Personal report	Occurs when a guest registration application approved by admin
Annual Report	General term describing the annual report of KADA	Organization report	Occurs when admin upload annual report pdf file to system

3.3.2 Identifying Relationship Types

Entity name	Multiplicity	Relationship	Entity Name	Multiplicity
Admin	1..1	update	Annual Report	1.. *
	1..1	approve	Members	1.. *
	1..1	approve	Loans	1.. *
Guest	1..1	become	Members	1..1
	1.. *	check	Annual Report	1.. *
Members	1..1	have	Savings	1..1
	1..1	have	Family	0.. *
	1..1	have	Staff Info	1..1
	1..1	apply	Loans	1.. *
	1..1	check	Individual Report	1..1
	1.. *	check	Annual Report	1.. *
Savings	1..1	include	Transaction	1.. *
Loans	1..1	have	Guarantor	[2]
	1.. *	have	LoanTypes	1..1
	1.. *	have	Banks	1..1

3.3.3 Attributes Description

Entity Name	Attributes	Data Type & Length	Description	Nulls	Multi-valued	Example
Guest	guest_id (PK)	int	Uniquely identifies each guest	No	No	1
	guestPassword	varchar(255)	Account password for login	No	No	\$sf@142
	guestUsername	varchar(255)	Guests' full name	No	No	Ali
	guestEmail	varchar(255)	Account email for login	No	No	staff@gmail.com
	member_id (PK)	int	Uniquely identifies each member	No	No	5
	name	varchar(255)	Members' full name	No	No	Ahmad
	ic	int	Members' identification card number	No	No	981119-01-8732
	email	varchar(255)	Account email for receive notification	No	No	ahmad@gmail.com
	sex	varchar(255)	Members' gender	No	No	Lelaki
	phone	varchar(255)	Members' phone number	No	No	011-16596391
	DOB	date	Members' birth date	No	No	19/11/1998

Members	noPF	int	Members' personal profile	No	No	6
	address	text	Members' address	No	No	No 128 Taman U
	status	ENUM	Indicates the application state	No	No	Johor
	city	text	City where the member is located	No	No	Muar
	postcode	int	Postcode where the member is located	No	No	84000
	state	text	State where the member is located	No	No	Kelantan
	office_address	text	Office address of member	No	No	Peti Surat 127
	office_city	text	Office city of member	No	No	Bandar Kota Bharu
	office_postcode	int	Office postcode of member	No	No	15710
	time	date	The date of submission application	No	No	05/1/2025
	familyNo	int	Family record of members	Yes	Yes	2
	staffNo	int	Staff number of member	No	No	4672

Admin	admin_id (PK)	int	Uniquely identifies each admin	No	No	1
	adminPassword	varchar(255)	Account password for admin login	No	No	\$gf@142@
	adminUsername	varchar(255)	Admins' full name	No	No	Ahmad
	adminEmail	varchar(255)	Account email for admin login	No	No	admin@gmail.com
Transaction	Transaction_id (PK)	int	Uniquely identified transaction record	No	No	6
	transactionType	ENUM	Type of transaction made	No	Yes	Loan repay , Deposit
	time	date	Date of transaction made	No	No	19/1/2025
	Savings_id (FK)	int	Indicated to the specific savings account info	No	No	7
	savings_id (PK)	int	Uniquely identified savings record	No	No	56
	savingAmount	float	Amount of the member savings	No	No	3000
	dividendRate	float	Yearly dividend rate	No	No	0.03

Savings	YuranMasuk	float	Amount of members' Yuran Masuk	No	No	2550
	ModalSyer	float	Amount of members' Modal Syer	No	No	1480
	ModalYuran	float	Amount of members' Modal Yuran	No	No	1320
	WangDepositAnggota	float	Amount of members' deposite	No	No	2550.50
	Al-Abrar	float	Amount of members' Al-Albrar	No	No	1000.50
	SimpananTetap	float	Amount of members' fixed deposit	No	No	6000.50
	amount	float	Total amount of the savings	No	No	20000
	member_id (FK1)	int	Indicated to the member who own the savings record	No	No	6
Family	FamilyNo (PK)	int	Uniquely identified family record of member	No	No	4
	name	varchar(255)	Members' family name	No	No	Ismail
	IC	varchar(255)	Members' family ic	No	No	931118-01-1236
	relationship	varchar(255)	Relationship between the	No	No	Adik lelaki

			family and member			
Staff Info	staffNo	int	Uniquely identified each staff	No	No	9824
	position	varchar(255)	Staffs' position in KADA	No	No	perkhidmatan pelanggan
	gred	varchar(255)	Staffs' grade in KADA	No	No	M40
Guarantor	GuarantorNo (PK)	int	Uniquely identified guarantor of member	No	No	7
	name	varchar(255)	Guarantors' name	No	No	Siti
	ic	varchar(255)	Guarantors' IC	No	No	991231-01-9823
	staff_id	int	Guarantors' staff ID in KADA	No	No	7
	phone	varchar(255)	Guarantors' phone number	No	No	019-7113981
	application_id (PK)	int	Uniquely identified each application record	No	No	6
	date_apply	date	Submission date of application	No	No	20/1/2025
	loan_amount	float	Amount of loan to apply	No	No	2000.50
	interest_rate	float	Interest rate for loan	No	No	0.05

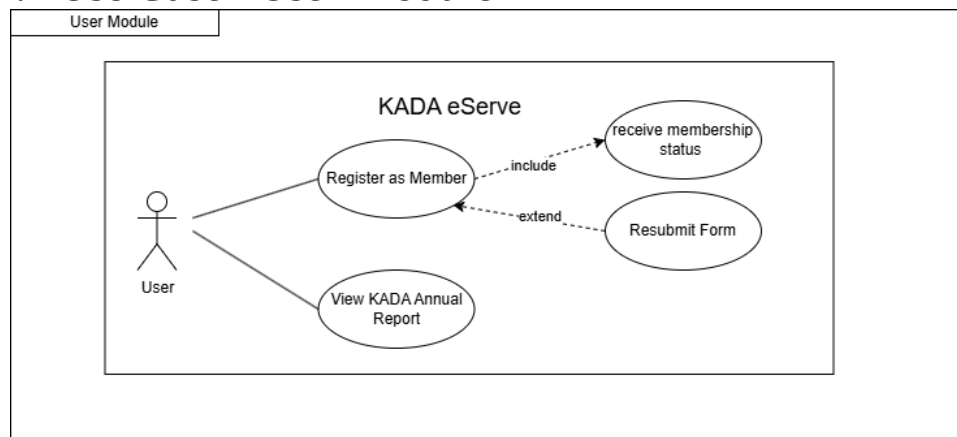
Loans	monthly_repayment	float	Monthly repayment of member	No	No	200
	monthly_gloss_salary	float	Monthly gloss salary of member	No	No	4000
	monthly_net_salary	float	Monthly net salary of member	No	No	3500
	loan_period	date	Period of loan apply	No	No	18/2/2028
	status	ENUM	Status of the application	No	No	approved
	time	date	Submission date of the application	No	No	19/1/2025
	noPF	int	Indicated to the personal profile of member	No	No	4
	member_id (FK 1)	int	Indicated to member ID	No	No	8
	loan_type_id (FK 2)	int	indicated to type of loan member apply	No	No	4
LoanTypes	bank_id (FK 3)	int	Indicated to bank record of member	No	No	6
	loan_type_id (PK)	int	Uniquely identified the type of loan	No	No	5
	loan_type	varchar(255)	Type of loan to be applied	No	No	Simpanan Tetap

Banks	bank_id (PK)	int	Uniquely identified each members' bank info	No	No	8
	bank_name	varchar(255)	Members' bank name	No	No	Maybank
	bank_account	int	Members' bank account number	No	No	123456789000
Individual Report	individualReport_id (PK)	int	Uniquely identified each members' individual report	No	No	8
	member_id (FK)	int	Indicated to which member own the report	No	No	7
	loan_id (FK)	int	Indicated to the loan application that the member made	No	No	2
	savings_id (FK)	int	Indicated to the savings record of the member	No	No	5
Annual Report	annualReport_id (PK)	int	Uniquely identified each annual report uploaded	No	No	4
	filePDF	BLOB ?	Annual report in PDF file	Yes	No	Binary file

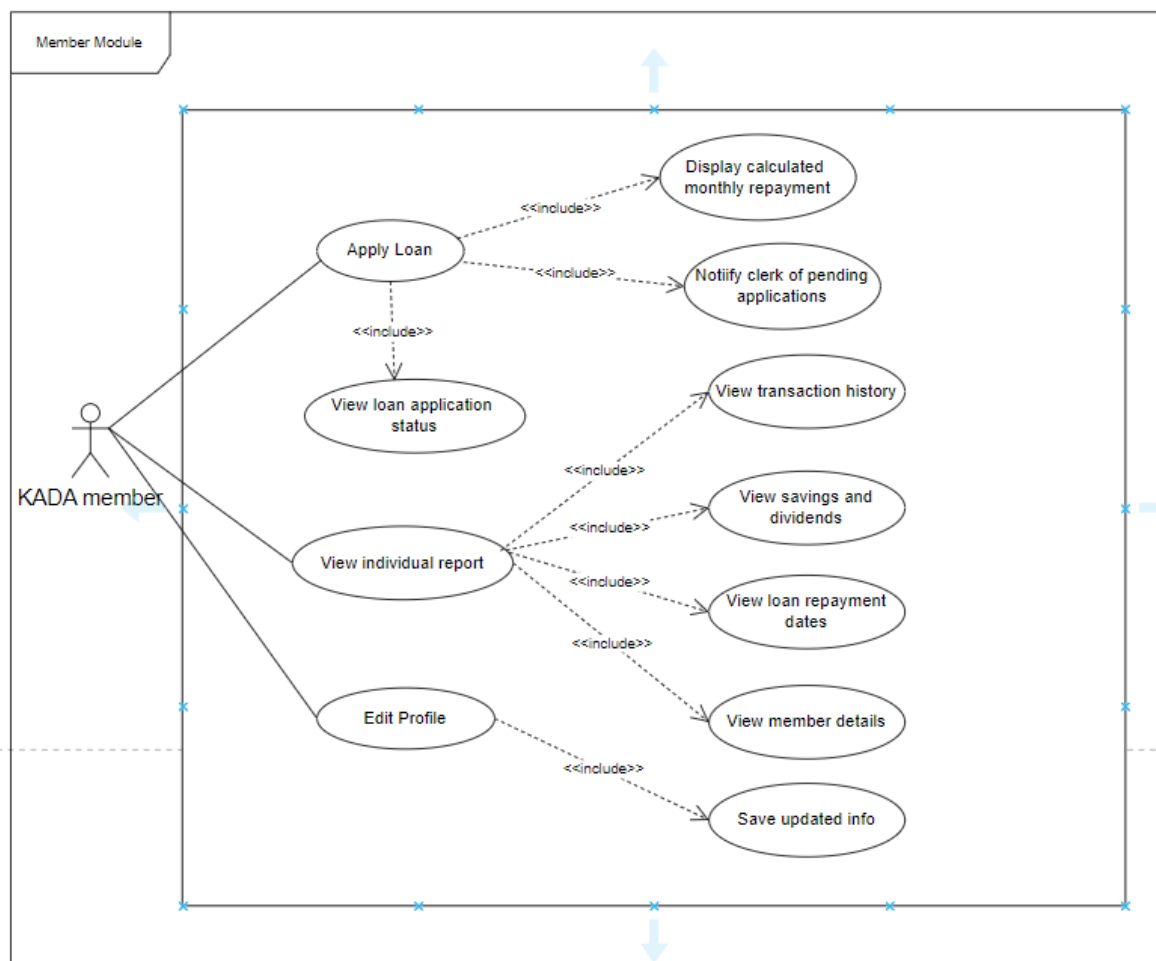
4. Appendices

4.1 Use Case Diagram for the system

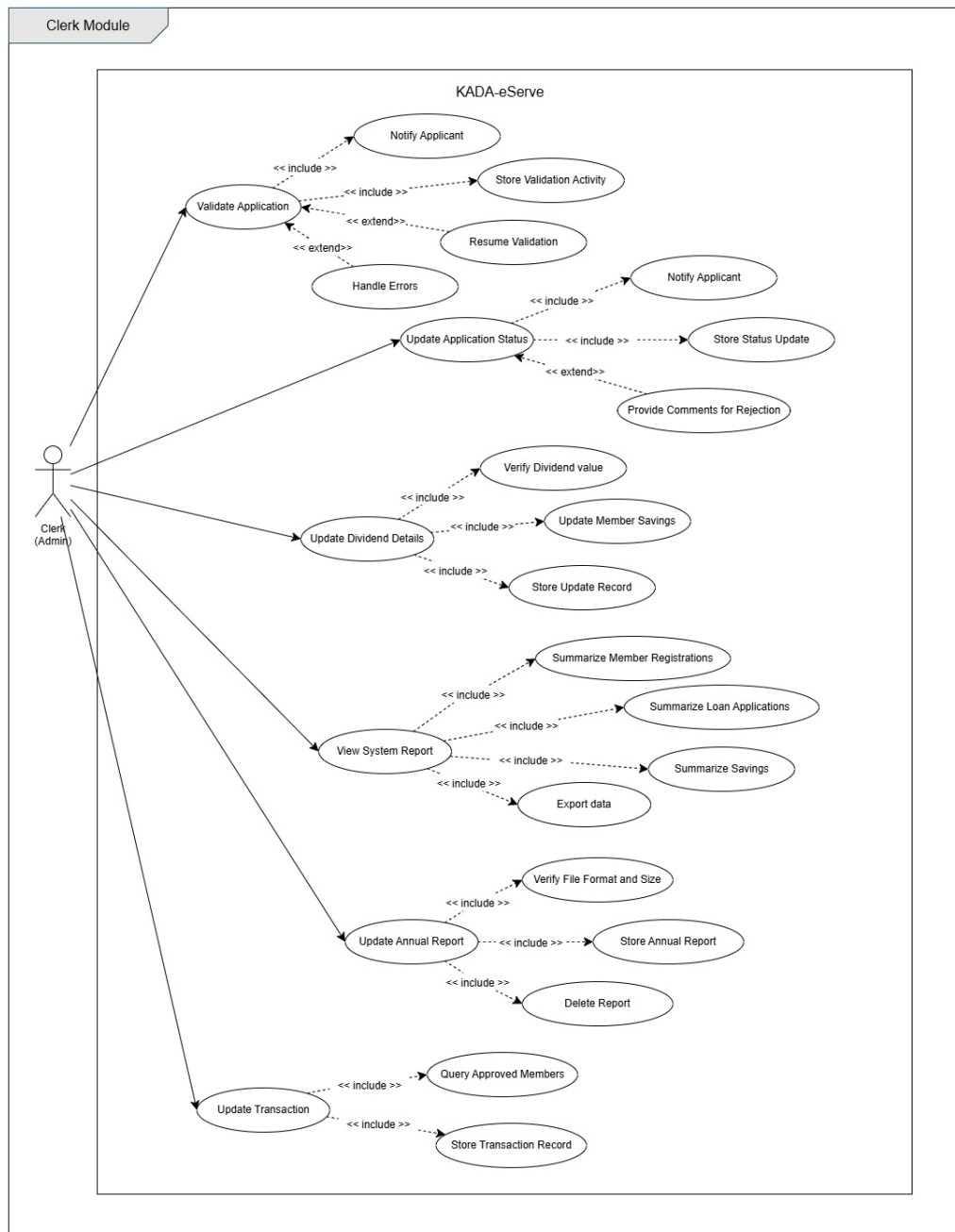
4.1.1 Use Case <User Module>



4.1.2 Use Case <Member Module>

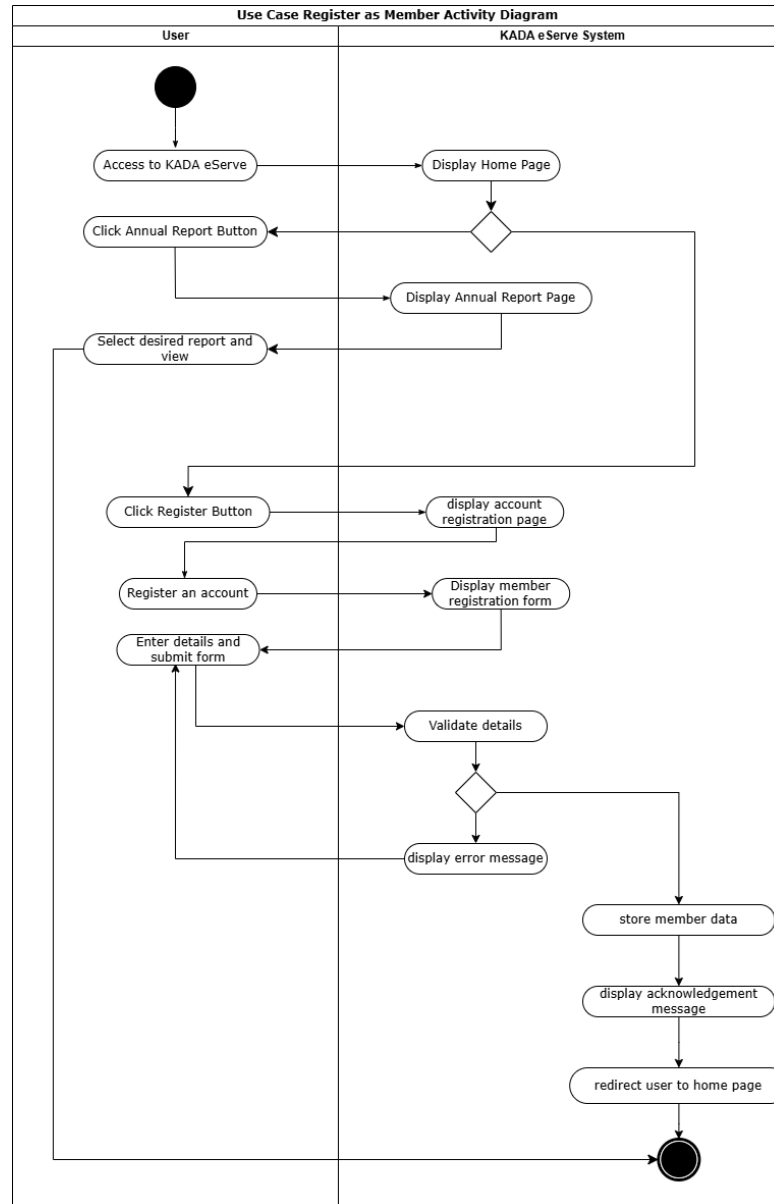


4.1.3 Use Case <Clerk Module>

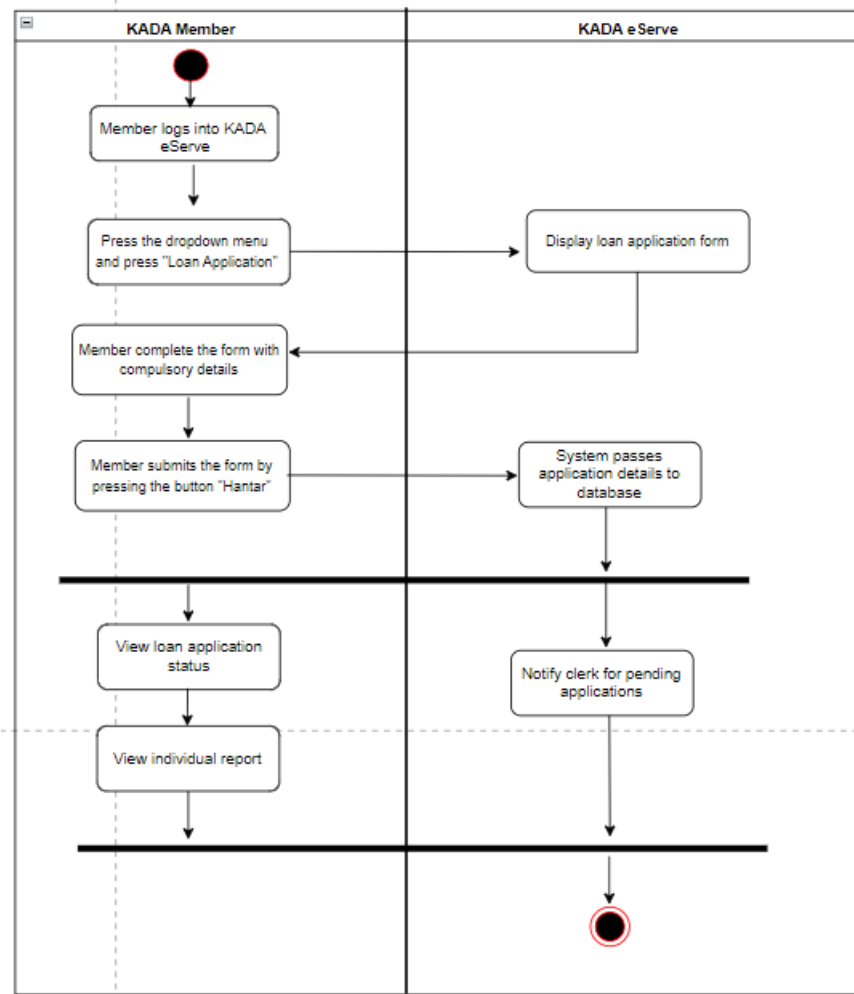


4.2 Activity Diagrams for every use case identified

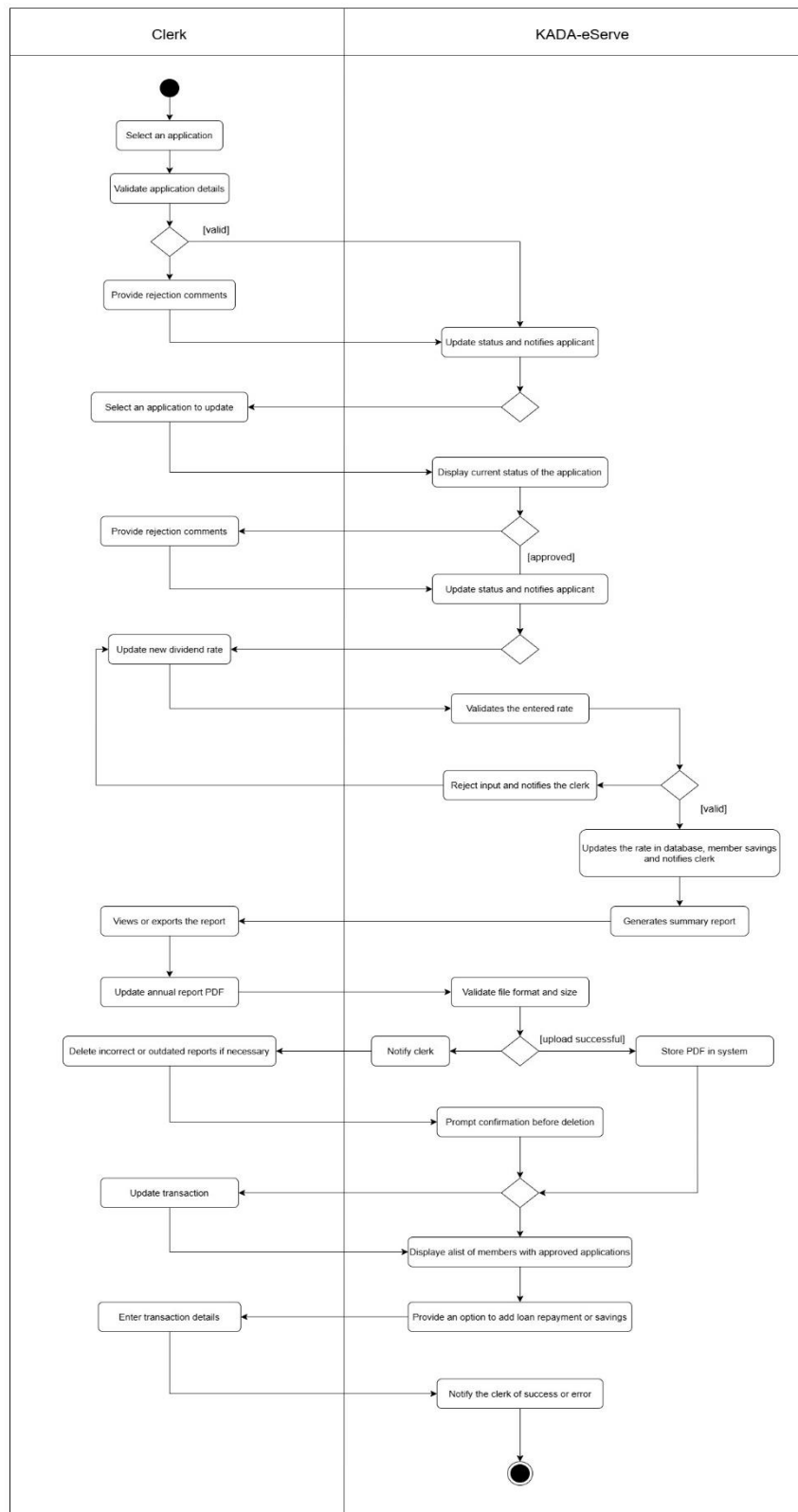
4.2.1 Module <User Module>



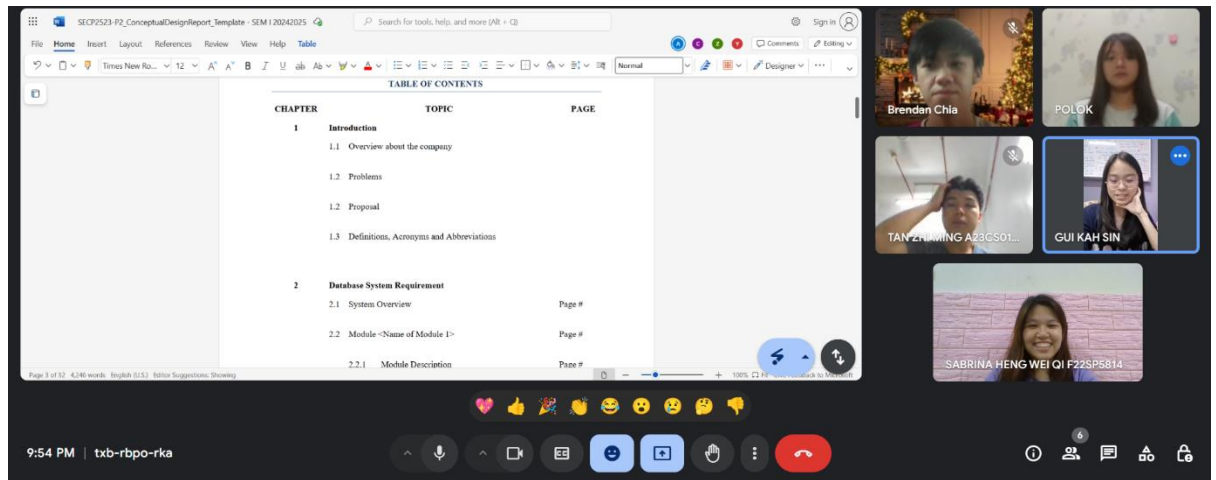
4.2.2 Module <Member Module>



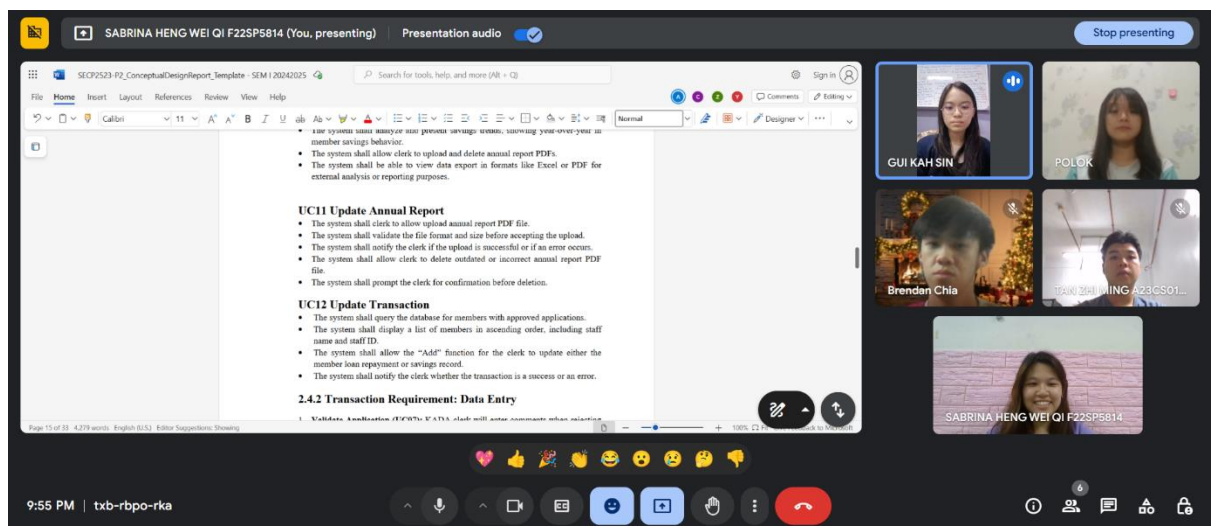
4.2.3 Module <Clerk Module>



4.3 Meeting Log



We start our meeting at 2:30 p.m. on 28 December 2024. In this meeting, the leader assign parts to each member and discussion about the second phase of the database project. Everyone attends the meeting punctually and having a smooth discussion time. Each member feels satisfied for the way of assign for their project parts. Lastly, this meeting is to have a brief discussion about the project and exchanging ideas about the ways to perform successfully in this project.



This meeting be conducted at 10 p.m. on 13 January 2025 and the purpose of the meeting is to finalize each part of the project and discuss about the problem faced during the project. Each member updates their progress and speak out about issues found during this documentation. In the end of the meeting, we solved all of the problems and proceed to complete the report.